



the heat. With more current needed by the load, transistor T1 conducts more and the current bypasses the LM317.

LED1 glows when 230V AC supply is on. VR1 is used to vary the output voltages according to your requirement. The voltage is shown on DPM.

Construction and testing

An actual-size, single-side PCB for the 3A variable power supply is shown in Fig. 4 and its component layout in Fig. 5. After assembling the circuit on PCB, enclose it in a suitable cabinet. Do not forget to use suitable heat-sinks for transistors TIP127, IC LM317 and IC LM7809.

Connect +12V power supply to exhaust fan from CON2 provided in the PCB. Also connect VR1 and LED1 via 2-pin wires from PCB to front panel.

PARTS LIST

Semiconductors:

- IC1 - LM317 adjustable voltage regulator
- IC2 - LM7809
- T1 - TIP127 PNP power transistor

- LED1 - 5mm LED
- D1-D4 - 1N5408 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 22-ohm, 2-watt
- R2 - 1.5-kilo-ohm
- R3 - 270-ohm
- R4 - 100-ohm
- VR1 - 4.7-kilo-ohm pot (potentiometer)

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 470 μ F, 25V electrolytic
- C3 - 0.1 μ F ceramic disc

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 4A secondary transformer
- S1 - On/off switch
- CON1 (for X1), CON2, CON4 - 2-pin terminal connector
- CON3 - 3-pin terminal connector
- DPM - Digital DC voltmeter
- 12V small exhaust fan (big size)
- Heat-sink for TIP127
- Heat-sinks for LM317 and LM7809

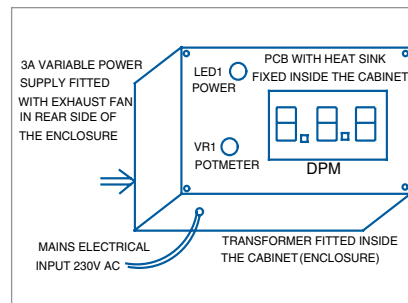


Fig. 6: Proposed layout of the parts in cabinet

Fix a small exhaust fan just behind the heat sink of TIP127 to get proper output. Also provide a window on the rear side of the cabinet for the exhaust fan's air outlet. Fix LED1, pot VR1, and DPM on the front side, and CON1 and CON2 at rear side of the cabinet. A placement of the parts in the cabinet is shown in Fig. 6. **EFY**

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